

Schema Design and Data Cleaning Rationale

ALP Software Case Study, Summit Ridge DataHub

Purpose

For this case study I treated the files like a first client export. My goal was to turn three messy spreadsheets into a clean Supabase database that a team could query and maintain. I kept the raw files, wrote repeatable cleaning code, loaded the cleaned records, and generated a one page HOA profile from the cleaned data.

I did not try to hide messy values. When a value was missing, partial, duplicated, or uncertain, I either normalized it with a clear rule or stored a review note in the data quality flags table.

Schema Choices

The associations table is the central HOA record. It stores the HOA code, name, city, state, unit count, monthly dues, fiscal year end, reserve balance, reserve study status, and board email with proper data types.

The board members table belongs to associations through a foreign key. I kept board service scoped to an association because the file does not prove that repeated names across different HOAs are the same person.

The vendors table stores one cleaned company row per real vendor. The association vendors table is the bridge between HOAs and vendors, which supports vendors serving several associations and associations using several vendors.

The vendor aliases table keeps source names that were merged during cleanup. The data quality flags table records the values I normalized, retained for review, or left blank because inventing an answer would be worse than showing the gap.

Cleaning Decisions

I normalized state values such as CA, ca, Calif., and California to CA. I parsed monthly dues and reserve balances into numeric values. Blanks, N A, and TBD became null when there was no trustworthy value.

Reserve study dates with only month and year were stored as the first day of that month, and I kept a precision field so the database still shows that the original source was not exact.

I removed the exact duplicate board row for Alex Tanaka. Missing board emails and open term dates were kept blank and flagged for review.

For vendors, I deduplicated by email first, then by phone and trade, then by normalized name and trade. That merged Bright Path Landscaping with Bright Path Landscaping LLC, Coastline Plumbing duplicate rows, and Reliable Electric with Reliable Electrical Services.

Vendors with no HOA relationship were not deleted. I kept them as prospects because that is useful operational data for a management company.

What I Did

I created the normalized schema, wrote the cleaning and seed script, generated the Supabase reset file, loaded the data into Supabase, and verified the final counts. The loaded database has 12 associations, 52 board member rows, 10 vendors, and 13 vendor links.

I also built the one page generator and generated an example profile for SR 04, The Beacon Pointe Association. I checked the output visually and fixed the layout so long emails did not overlap other fields.

My Judgment and Review

The vendor cleanup required the most judgment. I did not merge companies only because their names looked similar. I used matching email and phone values as stronger evidence, then preserved every source name as an alias so the original intake history was still visible.

I also chose not to replace unusual values just because they looked wrong. For example, I kept the zero monthly dues value and flagged it for review. I left unknown reserve balances and missing unit counts blank because a reviewer should be able to tell the difference between a confirmed zero and an unknown value.

After the load, I checked the database directly in Supabase. I verified the row counts, opened the tables, and confirmed that the relationships matched the cleaned files. I also generated the SR 04 profile and inspected the visual result before treating the work as complete.

If this were moving into production, my next step would be to add a simple review screen for the data quality flags so staff could confirm or correct uncertain records without editing SQL.

AI Use

I used AI as a working assistant while building the project. It helped me plan the schema, draft scripts, improve wording, and shape the one page generator.

I reviewed the work myself, corrected generated SQL after Supabase showed an error, reran the load, checked the row counts, inspected the HTML output, and made the final decisions about what to normalize and what to flag.